

LonghornSilicon

Cross-Block Architectural Conflict Register

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Abstract

A standing register of design tensions between the four blocks of the LonghornSilicon LLM inference accelerator: ACU (Attention Compute Unit, block 1), KVCE (KV Cache Engine, block 2), TIU (Token Importance Unit, block 3), and MHC (Memory Hierarchy Controller, block 4). Each entry catalogues a specific incompatibility, specification gap, or design choice that requires coordinated resolution across more than one block. The register is structured to be appended-to as integration work progresses, not rewritten each iteration. The companion document *KVCE* $\times$ *ACU Integration Audit* (kvce-acu-audit-0.2) covers the empirical measurement of items that have been quantified; this document is broader and catalogues items not yet measured.

1. Scope and conventions

The register covers conflicts and design tensions *between* LonghornSilicon blocks, not bugs within a single block (those are filed as GitHub issues against the respective block’s repository). A conflict warrants an entry here when:

- It cannot be resolved in any single block’s RTL or reference model alone.
- It requires a design decision that spans at least two of {ACU, KVCE, TIU, MHC, compiler}.
- It is not yet captured in a stable interface specification.

Each entry uses the following fields:

Severity

high = blocks integrated correctness; *medium* = degrades quality or limits configuration but chip runs; *low* = inefficiency or design-overhead issue.

Status

open = needs decision; *partially resolved* = discussed and tracked but not concluded; *resolved* = decision made and implemented in some interface; *deferred* = recognised but parked beyond the current tape-out target.

Affected blocks

which of {ACU, KVCE, TIU, MHC, compiler} are touched.

Resolution path

short description of how the conflict can be closed and what the candidate options are.

2. Summary table

#	Conflict	Severity	Status	Affected
1	Q4.12 input range vs. raw activation magnitudes	high	open	ACU, KVCE
2	FP16 routing depends on full-precision V available	high	partially resolved	ACU, KVCE
3	int16 Q4.12 $\leftrightarrow$ int8/fp16 bridge unspecified	high	open	ACU, KVCE
4	GQA convention not in KVCE ISA	medium	open	KVCE, compiler
5	TIU cold-start eviction policy undecided	medium	open	TIU, ACU, MHC
6	SRAM_DEPTH default of 16 below realistic hot-tier need	medium	partially resolved	KVCE, MHC
7	Eviction granularity vs. tile size	medium	deferred	KVCE, MHC
8	K (288b) vs. V (208b) word width waste	low	deferred	KVCE
9	eDRAM viability at 16FFC	medium	deferred	MHC, foundry
10	Lloyd–Max centroids vs. early-layer activation distribution	medium	open	KVCE
11	End-to-end accuracy on real LLM not measured	high	open	ACU, KVCE, integration
12	TurboQuant+ <code>turbo4</code> noise floor obviates PC FP16 path	informational	resolved	ACU, KVCE

The remainder of this document expands each entry.

3. Conflict 1 — Q4.12 input range vs. raw activation magnitudes

Severity: high. Status: open. Affected: ACU, KVCE.

KVCE accepts vectors as 64 lanes of int16 Q4.12 fixed-point, range ± 8.0 at resolution 2^{-12} . Real Qwen2-0.5B post-k_proj key activations measured in this work have absolute maximum approximately ± 152 at the layers sampled. A naive FP16 $\rightarrow$ Q4.12 conversion at the boundary clips approximately 95% of K ’s magnitude information before any compression takes place. The KVCE algorithm internally normalises the vector by its L2 norm before quantization, so the clip is unrecoverable: the post-clip vector has the wrong direction, and no downstream step in KVCE can restore it.

Resolution path. Three candidates, none decided:

1. Widen the KVCE input format (e.g. Q8.8 with ± 128 range, or Q12.4 with ± 2048). Breaks current RTL bit widths and all existing reference-model tests.

2. Add a pre-normalisation stage between the ACU’s `k_proj` output and the KVCE input port. The stage extracts a per-vector scale factor (similar to KVCE’s internal norm metadata but at a wider range) and rescales the input into Q4.12-representable range. Requires a new sub-block specification.
3. Document Q4.12 as the input contract and require upstream software / firmware to normalise activations into this range before streaming. Pushes the problem onto the compiler/runtime.

Status of tracking. Filed as `kv-cache-engine#2` (see also `kvce_acu_integration_audit.tex` §5(a)).

4. Conflict 2 — FP16 routing depends on full-precision V

Severity: high. Status: partially resolved. Affected: ACU, KVCE.

The precision controller’s value proposition is per-tile FP16 routing for “peaked” tiles where small attention-weight dimensions on high-mass key positions are necessary to preserve output accuracy. This argument was measured against V supplied at FP16 (i.e. no KV cache in the loop). When the chip integrates the KVCE, V is already lossy after ~ 3 -bit PolarQuant compression by the time it reaches the ACU’s FP16 SV matmul. The integration audit (`kvce-acu-audit-0.2`) measures this empirically: the FP16 vs. INT8 distinction is $\sim 1,800\times$ smaller than the KVCE-induced rMSE on the `turbo4` configuration.

The conflict is partially resolved in that the empirical answer is in hand — the PC’s FP16 path provides no measurable benefit at `turbo4`. It is not fully resolved because the design question remains: do we (i) retain the FP16 SV datapath knowing it is under-utilised at `turbo4` but valuable at higher-precision modes (`turbo8`, `turbo16`); (ii) simplify the ACU to INT8-only SV under the current configuration; (iii) commit to a higher-precision KVCE mode that restores the FP16 path’s value?

Resolution path. Decision deferred until either the chip’s target KVCE compression mode is locked, or higher-precision KVCE modes are implemented and re-measured. The ACU FP16 datapath remains in the design pending that decision.

5. Conflict 3 — int16 Q4.12 $\leftrightarrow$ int8/fp16 bridge unspecified

Severity: high. Status: open. Affected: ACU, KVCE.

KVCE decompresses each K or V vector to 64 lanes of int16 Q4.12. The ACU’s MAC array accepts either INT8 (for the INT8 path) or fp32-at-fp16-precision (for the FP16 path). The conversion unit that bridges these formats does not exist in any LonghornSilicon specification, RTL, or reference model. It currently lives implicitly inside “kernel software”.

The bridge has at least two distinct sub-cases:

- Q4.12 $\rightarrow$ int8: per-tile symmetric quantisation (`max_abs` scan, divide by 127, round). Used on the INT8 SV path.
- Q4.12 $\rightarrow$ float32-rounded-to-fp16: straightforward multiply by 2^{-12} followed by IEEE-754 binary16 rounding. Used on the FP16 path.

The bridge also has open questions about who picks the precision (driven by the PC decision; routed by the bridge itself; or routed upstream of the bridge); about where the per-tile scale lives (sent alongside the decision; recomputed on every tile); and about pipeline latency.

Resolution path. Specify the bridge as either (a) an ACU input-stage sub-block (preferred — the precision-routing logic naturally lives near the ACU), (b) a KVCE output-stage sub-block (less preferred — KVCE shouldn’t need to know about ACU’s INT8/FP16 routing), or (c) its own block between ACU and KVCE (over-architected for the role). Then write its RTL, reference model, and ISA at the same level of formality as the existing block specs.

6. Conflict 4 — GQA convention not in KVCE ISA

Severity: medium. Status: open. Affected: KVCE, compiler.

Qwen2-class models use grouped-query attention (GQA): for example, Qwen2-0.5B has 14 query heads and 2 key/value heads in a 7:1 ratio. KVCE’s ISA treats vectors as anonymous 64-D entities with no `head` field on the AXI-Stream interface (only `tuser` to select K vs. V path). The spec does not state whether:

- A separate KVCE instance is required per KV head, or
- One KVCE instance multiplexes the KV heads through address-space partitioning, or
- The compiler is expected to flatten heads into the address space, treating the block as head-agnostic.

Resolution path. Add a `KV_HEAD` field to the ISA register file, or document explicitly that the block is head-agnostic and the compiler is responsible for head-indexed addressing. The address-space approach is the cheapest in silicon but ties the compiler tightly to the storage layout.

7. Conflict 5 — TIU cold-start eviction policy undecided

Severity: medium. Status: open. Affected: TIU, ACU, MHC.

The Token Importance Unit (block 3) accumulates per-token attention weight over decoding steps and uses the accumulator as the eviction signal. In the first ~ 50 –100 decode steps, accumulator scores are roughly equal across tokens because the running sum has not yet differentiated them. If the on-chip working set exceeds SRAM capacity during this window, eviction by lowest-accumulator is essentially random. This may cause the loss of tokens that become important later (“heavy hitters”).

The conflict is not internal to the TIU: the eviction signal flows to the KVCE (which advertises `evict_needed/evict_addr`) and the MHC (which spills/refills from off-chip DRAM). The choice of cold-start policy affects all three.

Resolution path. Three published approaches available: (i) H2O’s deferred-eviction-until-window-fills; (ii) StreamingLLM’s pinned attention-sink tokens; (iii) ForesightKV’s learned-prior pre-seeding of the accumulator. Option (iii) is most principled and slots into the TIU’s planned interface without changing the eviction comparator — the accumulator is simply non-zero at the start of decoding. ForesightKV adds a small offline-trained scorer that runs at prefill; its hardware footprint and

training pipeline are open questions. See external inbound from A. Singh (2026-05-20, GitHub abhi1432-boop/heavy-hitter-kv-cache) for context.

8. Conflict 6 — SRAM_DEPTH=16 below hot-tier need

Severity: medium. Status: partially resolved. Affected: KVCE, MHC.

The current KVCE SRAM_DEPTH default is 16 entries, which is the value that drives the OpenLane Sky130 signoff and area numbers in the KVCE paper. For an LLM-inference workload at $\text{seq_len} \geq 512$, the working set is much larger; even with aggressive eviction the hot tier needs hundreds to thousands of entries. The ISA range allows up to 4096, but no synthesis or signoff has been run at deployment-realistic depths.

Resolution path. Re-run Yosys synth and LibreLane Sky130 signoff with SRAM_DEPTH=1024 (or whatever the target is) to produce area and power numbers that reflect deployment. This is recorded in the KVCE roadmap. The conflict is partially resolved because the configurability exists; it remains open because the signoff numbers we cite do not reflect a deployable configuration.

9. Conflict 7 — Eviction granularity vs. ACU tile size

Severity: medium. Status: deferred. Affected: KVCE, MHC.

KVCE evicts at vector granularity: one address, one K or V vector spilled. The ACU reads at 64×64 tile granularity: 64 K vectors and 64 V vectors per attention tile. If the MHC has to fetch evicted vectors from DRAM one at a time on demand, the ACU's MAC pipeline stalls for many cycles per tile on a cache miss. Burst fetch is required.

Resolution path. Constraint on the MHC: per-tile fetch requests should be coalesced into single DRAM bursts (typically 8 or 16 vectors per burst at LPDDR4X). The TIU eviction policy should evict at tile-aligned granularity when possible (a "tile eviction list" rather than vector-by-vector). Detailed design deferred to MHC specification.

10. Conflict 8 — K vs. V word-width waste

Severity: low. Status: deferred. Affected: KVCE.

A compressed K entry is 288 bits; a compressed V entry is 208 bits. SRAM_WIDTH is set to $\max(288, 208) = 288$ bits, so each V entry wastes 80 bits (28% of the word). The total SRAM area overhead is meaningful at deployment-realistic depths but does not affect functional correctness.

Resolution path. Either (i) separate K and V SRAM macros (doubles the macro count, more efficient bit utilisation), or (ii) packed allocation scheme (more complex address generation, better bit utilisation), or (iii) accept the overhead as the cost of having a single uniform SRAM interface. Deferred to v0.2 of the KVCE block.

11. Conflict 9 — eDRAM viability at 16FFC

Severity: medium. Status: deferred. Affected: MHC, foundry.

The original chip block diagram describes the MHC as managing “L1 SRAM (hottest KV) + L2 eDRAM 8–32 MB + off-chip LPDDR5”. eDRAM at TSMC 16FFC is a specialty option, often requiring additional mask layers and may not be included in standard university program offerings (Muse Semi, TSMC University Program). If eDRAM is unavailable, the L2 tier defaults to high-density SRAM with reduced capacity per mm².

Resolution path. Confirm with the foundry (TSMC University Program directly, or via Muse Semi) which on-chip memory IPs are available at 16FFC for student tape-outs. If eDRAM is unavailable, revise the architecture to “all-SRAM on-chip” and accept the capacity reduction; size LPDDR bandwidth accordingly. This decision needs to land before MHC specification begins in earnest.

12. Conflict 10 — Lloyd-Max centroids vs. early-layer activations

Severity: medium. Status: open. Affected: KVCE.

The KVCE compression’s 8 Lloyd-Max centroids ($\pm 0.2451\sigma$, $\pm 0.7560\sigma$, $\pm 1.3439\sigma$, $\pm 2.1520\sigma$ with $\sigma = 1/\sqrt{d}$) are derived under the assumption that the Walsh–Hadamard-rotated, norm-normalised vector has coordinates drawn from $\mathcal{N}(0, 1/\sqrt{d})$. This holds for mid-network activations but breaks at layer 0 (and similar near-input layers) because the model’s residual stream has not yet been smoothed by accumulated LayerNorms. The integration audit measured a $16\times$ rMSE spread between layer 0 (2.567) and layer 12 (0.159).

Resolution path. Candidate fixes are per-layer centroid tables (small register file) or an adaptive turbo4-adaptive mode in the compression algorithm. Filed as `kv-cache-engine#3`; see `kvce_acu_integration_audit.tex` §5(b).

13. Conflict 11 — End-to-end accuracy not measured

Severity: high. Status: open. Affected: ACU, KVCE, integration.

The integration audit measures per-tile relative MSE against a dense FP16 attention baseline. The integrated rMSE of 0.36 is suggestive but does not directly bound model-output quality. A serious chip-level accuracy claim requires measurement of downstream metrics: perplexity on a held-out corpus, HellaSwag accuracy, or similar, with the full attention pipeline (ACU + KVCE round-trip + softmax + SV matmul) substituted for dense FP16 attention. This has not been done.

Resolution path. Run a held-out evaluation (perplexity on wikitext-2 or HellaSwag) against Qwen2-0.5B with attention substituted by the integrated ACU $\times$ KVCE pipeline, and compare to dense FP16. Filed as `kv-cache-engine#4`.

14. Conflict 12 — turbo4 noise floor obviates PC FP16 path

Severity: informational. Status: resolved (measured). Affected: ACU, KVCE.

At the current KVCE configuration (turbo4: 3-bit PQ + 1-bit QJL for K , 3-bit PQ for V), the per-vector reconstruction noise dominates the precision controller’s INT8/FP16 routing distinction. Quantified in `kvce_acu_integration_audit.tex` §5(c): KV rMSE $\sim 1,800\times$ the PC routing rMSE.

This is a design property of `turbo4`, not a defect. The PC's FP16 path would re-acquire value at higher-precision modes (`turbo8`, `turbo16`) where the per-vector noise drops. Listed here for completeness.

15. Closed items (for reference)

KVCE reference-model reconstruction bug

KVCE issue #1: `normalize()` used `coord_frac=12` instead of `norm_frac=8` for left shift, causing $16\times$ overscale; C++ truncation division differed from Python floor division. Fixed in `kv-cache-engine@9b1163a`; quality gate added at `sw/reference_model/test_reconstruction_quality.py`. **Closed.**

Change log

- **conflicts-0.1** (2026-05-22): initial register compiled from the original cross-block conflict scan (2026-05-15) plus residual items surfaced by the integration audit (2026-05-19, 2026-05-22) plus TIU cold-start item from inbound collaborator (2026-05-20).